



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

GRADELA

GRADient Evaluation & Learned Academic Performance Radar

End-to-End Predictive Modeling, Ensemble Benchmarking,
and Proactive Student Intervention Triage Pipeline.

Student Performance Prediction

BUSINESS INTELLIGENCE

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Executive Agenda

1. Problem Statement & Dataset Architecture

Challenges in academic retention and data schema

2. Exploratory Data Analysis (EDA)

Correlation landscape and multicollinearity checks.

3. Methodology & Engineering

Risk thresholding, median/mode imputation, and stratified partitioning.

4. Model Formulation & Ensemble Learning:

Benchmark comparison of Logistic Regression, Random Forest, and Gradient Boosting (GBM).

5. Model Diagnostics & Performance Evaluation

Confusion matrix statistics, ROC-AUC comparison, and calibration.

6. Model Interpretability & Classroom Rules

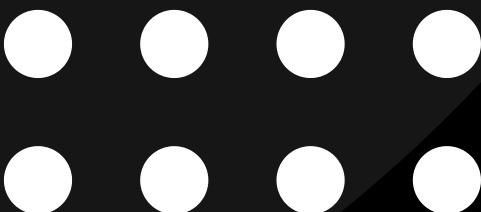
Feature importance ranking and white-box Decision Tree intervention paths.

7. Continuous Score Regression

Exact exam score forecasting (R^2 , RMSE, MAE).

8. Educator Triage System & Conclusion

Operational risk priority tiers and deployment recommendations.



Problem Statement & Dataset

THE PROBLEM

Late identification of struggling students leads to course failure and dropouts. Traditional reactive grading methods fail to provide early, actionable alerts.

THE GOAL

Build an early-alert machine learning pipeline (GRADELA) to classify student risk profiles and forecast exact marks before final assessments.

Source: Student_Performance_Factors_Dataset.csv

6,607

students in the total cohort

20

feature dimensions

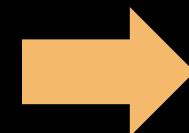
0–100

target: exam_score (continuous)



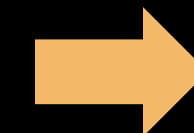
Raw student inputs

Behavioral & demographic features



GRADELA engine

Learned risk model



Early Warning Risk Alert



Continuous Score Prediction

Data Engineering & Methodology

Preprocessing & Imputation

- Standardized column names to snake_case
- Numerical gaps filled with the median (median(na.rm = TRUE))
- Categorical gaps labeled "Unknown" and encoded as factors

Risk Formulation (25th Percentile Rule)

- Cutoff threshold: final exam score ≤ 65.00
- Binarized target: High_Risk (≤ 65.00) vs. Normal (> 65.00)

Validation Strategy

- Stratified 80/20 train/test split (createDataPartition, seed 42)
- 5-fold cross-validation with ROC optimization (twoClassSummary)

CLASS DISTRIBUTION (n = 6,607)

High-Risk: 2,131 (32.25%)

Normal: 4,476 (67.75%)

Exploratory Data Analysis

Study volume & attendance

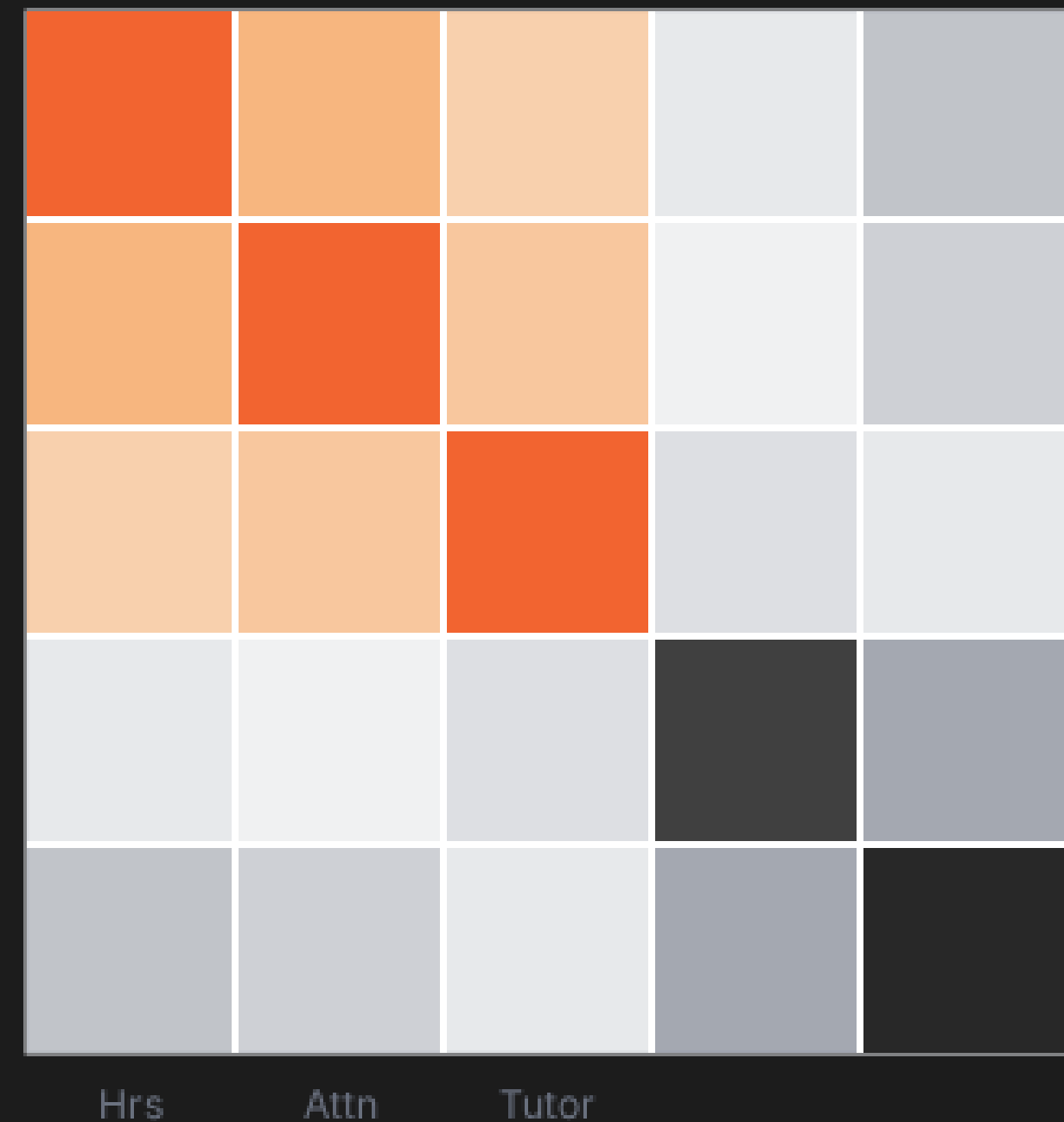
hours_studied and attendance show the strongest positive correlation with academic performance.

Support services

tutoring_sessions shows a notable positive association with scores.

Manageable collinearity

Predictors across behavioral and demographic variables confirm stable linear and tree-based model training.



Models & Ensemble Architecture

1

Logistic Regression (GLM)

Baseline

Generalized linear model with a binomial family link.

2

Random Forest Classifier

Bagging ensemble

100 de-correlated decision trees with out-of-bag validation and random feature subsets.

3

Gradient Boosting Machine (GBM)

Boosting ensemble

Sequentially boosted shallow decision stumps optimizing pseudo-residuals via gradient descent.



Hyperparameter tuning: repeated 5-fold cross-validation saving class probabilities for ROC-AUC optimization.

Classification Results

91.14%

Overall Accuracy

95% CI: 89.48%–92.62%

82.16%

Sensitivity

Recall for High Risk

95.42%

Specificity

Identifying Normal

89.51%

Precision (PPV)

Positive predictive value

91.83%

NPV

Negative predictive value

0.7929

Cohen's Kappa

Substantial reliability

88.79%

Balanced Accuracy

Mean of recall rates

67.75%

NIR baseline

p-value < 2.2×10^{-16}

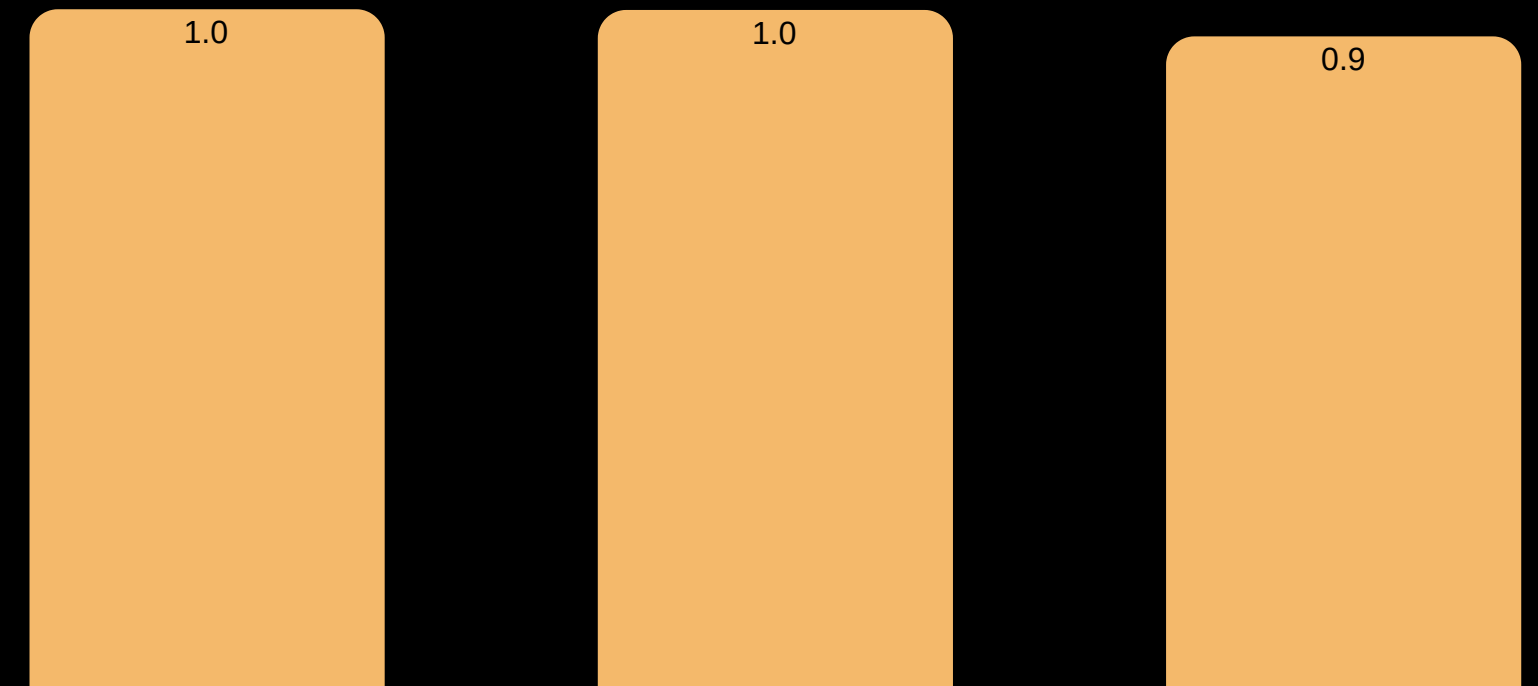
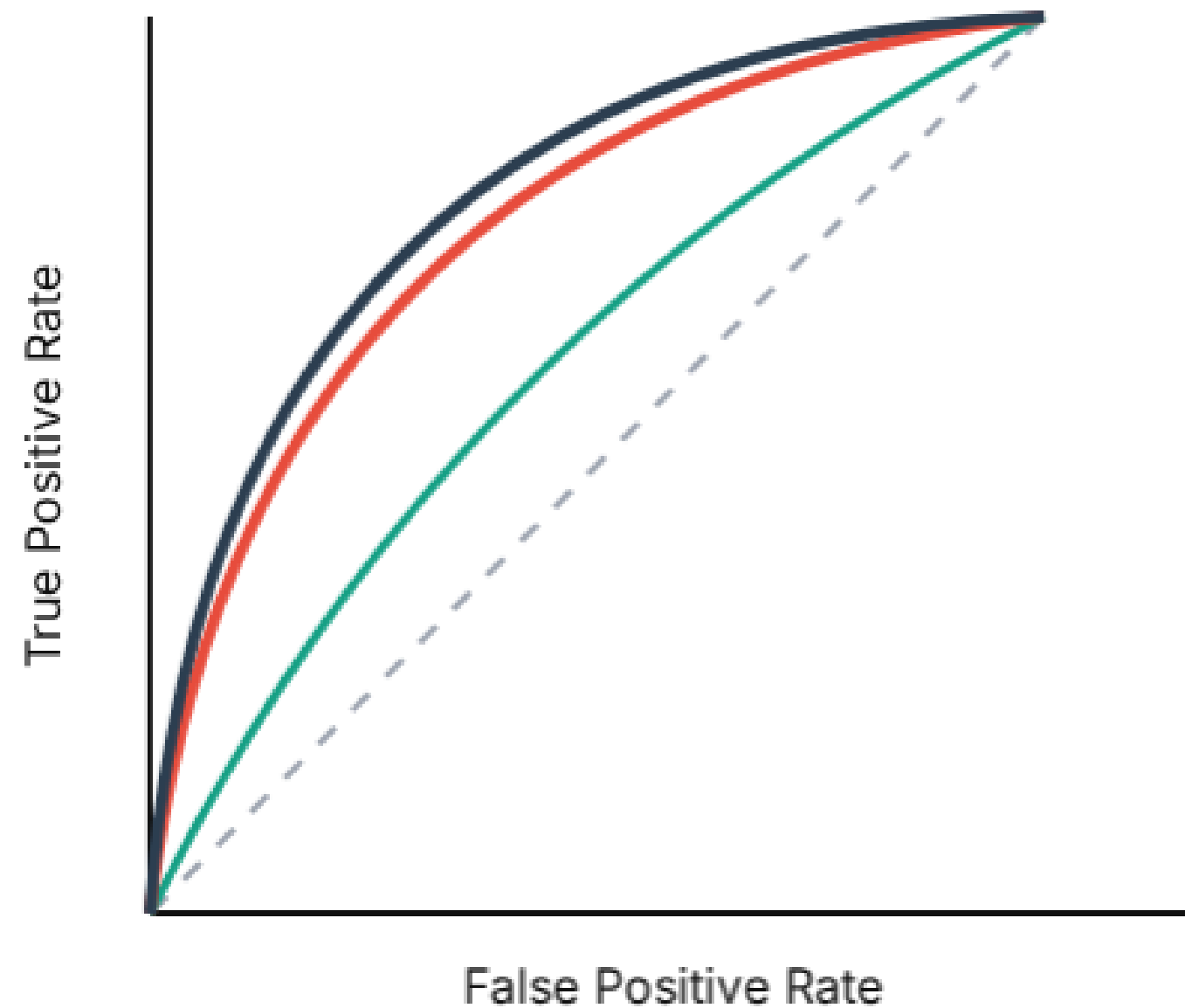
Random Forest classifier, hold-out test cohort. Accuracy beats the 67.75% no-information baseline by more than 23 points.

Validation Confusion Matrix

	Actual: High Risk	Actual: Normal
Predicted: High Risk	350 True Positives	41 False Positives
Predicted: Normal	76 False Negatives	854 True Negatives

High Risk correctly identified: 350. Normal correctly identified: 854. High Risk missed: 76.

Model Diagnostics: ROC-AUC



- Both ensembles significantly outperform the linear baseline by capturing complex non-linear factor interactions.
- Random Forest achieved the highest discriminative stability across classification thresholds.

Feature Importance

1

hours_studied

Leading primary predictor

2

attendance

Consistent top determinant

3

tutoring_sessions

Support services

SECONDARY ENVIRONMENTAL & BEHAVIORAL FACTORS

parental_involvement

access_to_resources

sleep

motivation

hours_studied

attendance

tutoring_sessions

motivation

sleep_hours

Key finding: risk is driven by actionable study habits and institutional engagement, not static demographics.

Classroom Intervention Rules

White-box splitting

rpart tree with $cp = 0.015$ turns ensemble predictions into simple, human-readable triage rules for counselors and teachers.

Clear trigger thresholds

Identifies the study-hour, attendance and tutoring-session values that trigger a high-risk classification.

No model queries needed

Non-technical staff can see why an intervention is required without querying the raw model.

Attendance < 75%

High Risk Flag

Hours Studied < 10

Continuous Score Forecasting

RANDOM FOREST REGRESSOR · 120 TREES

0.6322

R^2 · 63.2% of exam variance explained

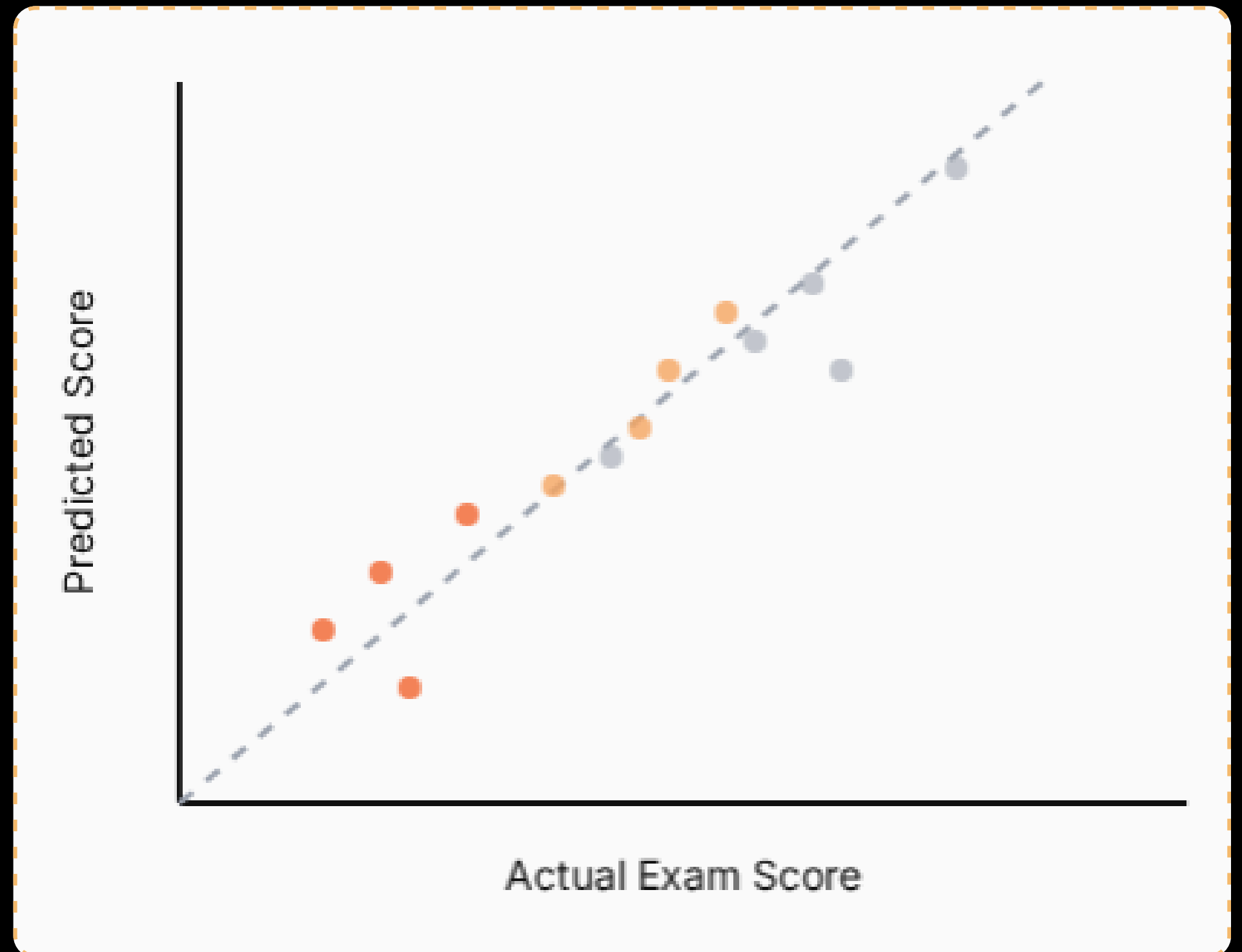
2.40

RMSE (marks)

1.08

MAE (marks)

- Predictions deviate by only ~1 mark on average across a 100-point scale.
- Dual utility: coarse-grained risk alerts plus fine-grained score forecasts.



Educator Triage System

Critical
Immediate action

≥ 75%

High Priority
Targeted outreach

50–74%

Moderate Watchlist
Monitor closely

25–49%

Low / On-Track
No action needed

< 25%

Tiers defined by predicted risk probability

391

students flagged for targeted academic support in the test cohort alone

TOP FLAGGED STUDENTS (NOTEBOOK OUTPUT)

Student	Risk	Tier	Hours	Attendance	Tutoring
Student 1	100%	Critical	15	70%	1
Student 2	100%	Critical	14	65%	1
Student 3	100%	Critical	17	61%	1
Student 4	100%	Critical	15	65%	1
Student 5	100%	Critical	11	63%	0

Auto-exported roster: `gradela_intervention_roster.csv`

Conclusions & Roadmap

PROJECT ACHIEVEMENTS



Developed a balanced, high-precision classifier achieving 91.14% accuracy and an AUC of 0.957.



Maintained high sensitivity (82.16%) so vulnerable students are reliably identified.



Delivered a production-ready intervention spreadsheet (gradela_intervention_roster.csv).

FUTURE WORK & DEPLOYMENT



Integrate into a live learning management system (LMS) dashboard via an R Shiny interface.



Incorporate periodic midterm updates for continuous, dynamic risk tracking.

91.14%

Accuracy

0.957

AUC

82.16%

Recall